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24ANDMORE: USING GENOME-WIDE SNPS TO INVESTIGATE THE GENETIC DIVERSITY OF IOWA WALLEYE (*SANDER VITREUS*) BROODSTOCK

An Abstract of a Thesis

Submitted

in Partial Fulfillment

of the Requirements for the Degree

Master of Science

Rachel Ann McDonnell

University of Northern Iowa

December 2021

ABSTRACT

The stocking of *Sander vitreus* in Iowa waterbodies is one of the primary tools used by the Iowa Department of Natural Resources (DNR) to ensure healthy populations for enjoyable fishing experiences. A substantial amount of time and money goes into this effort, so stocking efficiency is vital. Knowledge of genetic diversity contained in the broodstock increases this efficiency by having a better understanding of which strain will survive and contribute to the fishery in certain environments. The objective of this study is to investigate the genetic diversity of Iowa *S. vitreus* broodstock to identify genetic ancestry and quantify the genetic diversity of each strain contained in the broodstock. From this information, the creation of a strain-specific identification marker was explored. Rathbun Fish Hatchery's captive walleye broodstock contains two genetic strains: the lake-strain and the river-strain. Sampling of the 2019 captive broodstock at the Rathbun Fish Hatchery in Moravia, IA was conducted to genotype individuals and generate genome-wide single nucleotide polymorphism (SNP) datasets. Three datasets were generated; one to infer genetic variation between strain, the combined LR dataset, and two to infer genetic variation within strain, lake-only and river-only datasets. In total, 69 individuals were sampled and 11,764 SNPs were analyzed for the combined LR dataset. The datasets demonstrated the ability to detect fine-scale structure both between and within strains. The results of the analyses support a genetic differentiation between the two Iowa walleye strains; however, this variation is complex and subtle. Multiple genetic markers significantly associated with both lake- and river-strain were identified. The variation identified when the loci were compared between strains was not enough to

differentiate the strains and, therefore, create a stock identification tool. Based on these results, a simple strain-identification tool is not recommended. Instead, genotyping and generating more SNP datasets would not only allow for the unknown samples to be evaluated alongside the known individuals of this study, revealing genetic strain, but also would give further insight into the ancestry of Iowa walleye by revealing admixture and wild types that were not included in the present broodstock individuals.

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This Study by: Rachel Ann McDonnell

Entitled: 24andMore: Using Genome-Wide SNPs to Investigate the Genetic Diversity of Iowa Walleye (*Sander vitreus*) Broodstock

Has been approved as meeting the thesis requirements for the

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CHAPTER 1

INTRODUCTION

The walleye (*Sander vitreus*) is a popular freshwater sport fish across the United States. However, many inland waterbodies have limited to no natural reproduction due to both anthropogenic and natural factors, such as fishing pressure and predation (Hansen et al. 2011; Ivan et al. 2010). To mitigate low recruitment and support recreational fishing, the strategy of stocking has been implemented to maintain healthy populations (Johnson et. al 1996; Paragamian 1988). Stocking of walleye has been an important management tool used by many states and provinces for the last 70 years (Fenton et al. 1996, Halverson 2008).

Fish stocking is broadly defined as the practice of rearing a species in captivity and releasing it into a waterbody to supplement an existing population or to create a new population (Penne 2020). Stocked fish derive from broodstock, which is the population of the chosen species used for breeding. Rearing a species in captivity takes a significant amount of time and money; therefore, the survival of the stocked fish is imperative. However, the survivability of stocked fish to the adult population is often variable. When factors such as size of fish, location of entry, waterbody type, and fitness favorable to the target environment are considered, the chances of a successful addition to the adult population increase. Fitness, or the ability to adapt and reproduce, plays a key role in survival of stocked fish to the adult population (Orr 2009).

The fitness of a fish may be determined by genetic strain. Genetic variation of individuals and genetic diversity of the population give insight to the fitness capabilities

of the species. Higher genetic diversity allows a species or population to successfully combat negative survival and reproduction pressures, enhancing its fitness level (Zhongming et al. 2021). Conversely, releasing captive-born individuals into wild environments has been found to negatively affect genetic diversity and fitness level when genetic adaptation to captivity has a heritable basis (Willoughby and Christie 2019). The short-term and long-term effects of releasing captive-born individuals into wild environments is difficult to determine yet is necessary to consider in order to meet the goal of stocking fish that successfully contribute to the fishery.

Presently, walleye are found naturally in lakes and rivers throughout their range. One would assume that walleye sourced from rivers are better adapted to characteristics of river environments and walleye sourced from lakes are better adapted to characteristics of lake environments, due to containing heritable traits conducive to their respective environments. When stocked into a reservoir, which has some resemblance to lake characteristics and some resemblance to river characteristics, it is hypothesized that lake-strain fish would have a better chance of contributing to the adult population than river-strain fish in more lake-like reservoirs (i.e., those with rounder shape, lower turnover rate). Likewise, river-strain fish would have a better chance of contributing to the adult population in more river-like reservoirs (i.e., those with more dendritic shape, higher turnover rate). The genetic diversity of Iowa's walleye broodstock was investigated to understand the variation present within each strain, ultimately leading to the testing of this hypothesis. The knowledge of the genetic variation contained within each strain can be applied as stocking guidance and identification.

Figure 1. An adult walleye (*Sander vitreus*). Art by Maynard F. Reece.

Walleye Stocking and History

Sander vitreus is a native percid found in the inland waters of North America (Figure 1). They are a potamodromous fish native to the Mississippi River basin and the Mobile basin and have been stocked across most of North America for recreational fishing (Fuller and Neilson 2019). Reproduction of the species in the wild is naturally limited due to a single annual spawning event, which peaks over one to two weeks in spring. During spawning events, females move from deep winter habitats to shallower waters with rubble bottom and flowing water where the males are already waiting. Males move to spawning areas in early spring when water temperatures warm to just over freezing. On average, females lay between 12,500-55,000 eggs per pound of body weight (Wolfert 1969; Malison and Held 1996). Around 95% of the eggs will get fertilized, but survival quickly decreases due to predation, irregularly cool temperatures which delay hatching, heavy wind and strong currents which wash eggs ashore, fluctuation in water

levels, and muddy waters which coat eggs with silt (DeBoer and Pope 2015; Crane and Farrell 2013).

Iowa Walleye

Sander vitreus is naturally occurring in Iowa's glacial lakes and rivers but is also stocked to provide fishing in impounded lakes. The species is one of the most important culture products in the state. This is due to the combination of angler desire and limited natural reproduction in Iowa. Walleye continually rank high in angler catch and harvest in summer pool-wide and winter tailwater angler surveys. According to the 2018 Angler Survey, they are the most popular targeted sport fish species in Iowa (Responsive Management Plan 2019). Annual stockings of fry and fingerlings are vital to sustain and improve this important fishery. It is supported by a large hatchery culture and stocking program, which produces 160 million fry and over 405,000 fingerlings annually. These are then stocked for a total cost of $442,306.28 dollars according to the 2018 Culture Report from the Iowa Department of Natural Resources (DNR). The Iowa DNR have set their standard Walleye management goal of maintaining at least 3 adults/acre (Gelwicks et al. 2019). However, fisheries have had a difficult time achieving this goal due to frequent fry failures, high overwintering mortality, and high escapement.

Stocking has occurred in Iowa as early as 1880 with sourcing from the Ohio River (Paragamian 1988). Walleye sourced from out-of-state, having multiple genetic backgrounds, have also been used in Iowa for supplemental stocking. Rathbun Lake near Moravia, IA is the primary broodstock lake in the state and was established in 1969. It is a man-made reservoir, meaning the waterbody was created upon installation of the

Rathbun Dam, used for flood risk management, recreation, fish and wildlife management, water quality, water supply, and navigation (US Army Corps of Engineers).

The first fish stocked into Rathbun lake were cultured from Spirit Lake, IA and were coined as “lake-strain walleye”. Spirit Lake fish have a mixture of genetic backgrounds due to being stocked from multiple original sources, including from the Mississippi River (Paragamian 1977). Currently, lake-strain walleye originated from broodstock collected at Storm Lake, Lake Okoboji, Clear Lake, and Rathbun Reservoir. Even with the early introduction of Mississippi River fish in Spirit Lake, these lake-strain walleye may still be genetically different from “river-strain” walleye. River-strain walleye originate from sources along the Mississippi River including Fairport State Fish Hatchery, Cordova Power Plant Fish Hatchery, and Genoa National Fish Hatchery (Gelwicks et al. 2019). Today, the Rathbun Fish Hatchery captive broodstock contains both genetic strains.

Stocking Genetics

To properly manage or conserve a species, one must consider its genetic diversity, relatedness, and population structure. Stocking has been found to be the most successful when fitness of the donor population is considered and matched to the new environment (Everhart et al. 1975; Waltner 1988). Fitness is directly related to the genetics of the stock population due to genetic variability and its support against negative pressures. Fish with higher variability have a higher fitness and, therefore, success rate (Krueger and Dodson 1981; Zhongming et al. 2021). Yet, in the 1980s it was common to transfer walleye from one geographic region to another, even though not much was known about

the genetic components of the populations in each region (Murphy and Lee 1986). In 1986, Philipp et al. found that 17 out of 42 states genetically assess their fish stocks by comparison of hatchery strains and identification and preservation of native stocks. Further studies concluded that genetic strain of broodstock and its relationship with the target environment could significantly impact long-term survival rate and recruitment to fishable stock (Billington et al. 1992; Billington 1996; Page et al. 2017; Terre et al. 1995; Jennings et al. 1996; Logsdon et al. 2009). Today there are many projects underway to better understand the genetics of statewide broodstock programs (Haponski and Stepien 2016; Allen et al. 2018; Chen et al. 2020; Euclide et al. 2021).

Iowa Walleye Genetics

In a study from 1986 to 1989 three strains were stocked into Iowa rivers: lake-strain, Mississippi River-strain, and an Osage River, Missouri-strain to understand which strain may best contribute to the fishery. The study revealed that size-at-stocking influenced successful contribution and lake-strain had the least successful contribution (Paragamian and Kingery 1992). Another similar study comparing lake- and river-strains in Iowa found that the Mississippi River-strain "vastly outperformed" lake-strain in contributing to the river fishery, producing between 1.8 and 8.6 times more fish (Gelwicks 2001). These results greatly increased the demand for two-inch long, Mississippi River-strain walleye fingerlings from hatcheries throughout the state. However, the request is difficult to fill with limited hatchery space, so research is underway to compare intensively reared and extensively reared river-strain walleye. In June 2016, 61,000 river-strain marked fingerlings from Rathbun Fish Hatchery were

stocked in the Wapsipinicon, Maquoketa, Shell Rock and Cedar rivers and will continue to be sampled to determine survival and growth of each culture-type walleye (Gelwicks 2020). In 2020, another study was initiated to examine lake- and river-strain survival and contribution in reservoirs with a range of environmental conditions. The results will aid in guiding fingerling production and stocking methods in Iowa rivers.

Strategic stocking is necessary for an effective program. This involves accounting for not only general, but also strain-specific requirements for optimal fitness and survival. Stocking fish that lack the heritable traits favorable to survival in certain environments can result in ineffective stocking with higher costs and mortality rates. The investigation of genetic diversity in Iowa walleye broodstock will add substantially to the advancement of stocking, so the Iowa DNR can maintain healthy walleye populations.

Genetic Diversity of Broodstock

Information on genetic variation is paramount in the management of small populations maintained to serve as a source for population replacement or the establishment of new populations. Wild populations may be locally adapted to their environments (Jensen et al. 2008; Westley et al. 2013) and introgression of hatchery-reared fish could negatively impact such adaptations by introducing less-adaptive alleles (Reed et al. 2015; Lemopoulos et al. 2018). Evaluating the genetic differences between hatchery broodstock and native fish enhances successful replacement and/or establishment by preventing the loss of valuable data that can aid management decisions to conserve local genetic and morphological adaptedness (Kocovsky et al. 2013; Stepien et al. 2015). To account for the complex stocking history of Iowa walleye, single

nucleotide polymorphisms (SNPs) were obtained because of their ability to capture high numbers of sampled loci and, therefore, higher resolution of the data.

Significant Marker Identification

The ability to accurately identify the contribution of each stock to the fishery ensures that all stocks are being managed sustainably (Billington 1996; Vinther et al. 2004; Chen et al. 2020). Thus, if it is found that one strain of Iowa walleye outperforms the other in certain environments, production of the desired strain can be stocked with higher success rates if conservation of genetic variation is considered. Current stock identification methods have proved unreliable, but next generation sequencing has provided a method to create a tool for effective stock identification.

Study Objective

The objective of this study is to investigate the genetic diversity of Iowa walleye broodstock. The goal is to discover how diverse the lake-strain and river-strain broodstocks are by employing genome-wide SNP datasets. Analysis of these datasets will allow for identification of genetic ancestry and quantification of genetic diversity of each strain. This will allow better understanding of which strain may have more stocking success in certain environments. Stock identification methods will be explored to efficiently identify the strain that has the best survivability in Iowa reservoir environments.

CHAPTER 2

METHODS FOR RAD-SEQ

Restriction site-associated DNA sequencing (RAD-seq) is a next generation sequencing (NGS) method that yields a reduced representation of the target genome. Microsatellite datasets have been reliable in resolving the genetic diversity and structure, or patterns in genetic makeup, of populations but have limited ability in individual-level genotype evaluation (Lemopolous et al. 2018). The reduced representation of the genome, characterized by SNP datasets, consist of thousands of loci, allowing for high resolution and fine-scale delineation from low numbers of individual samples. An extreme example of this includes a simulation study that found a sample size as small as two individuals was able to obtain an accurate F_{ST} estimate due to analysis of more than 1500 SNPs (Nazareno et al. 2017). When compared to microsatellite datasets, SNP datasets can quantify relatedness and individual-level heterozygosity with higher probability and can better compare both strongly and weakly diverged populations, revealing ancestral patterns of genetic structure due to slower SNP mutation rates (Lemopolous et al. 2018; Andrews et al. 2016; Zhang and Hewitt 2003).

RAD-seq has made nearly all restriction sites of a particular restriction enzyme available for screening of SNPs, which is almost ten times more than traditional methods (Coates et al. 2009). The process first involves cutting the DNA with a restriction enzyme, allowing over sequencing of the nucleotides adjacent to restriction sites. The fragments of DNA, or RAD tags, then undergo massively parallel and multiplexed sample sequencing focused on the same subset of genomic regions across multiple

individuals. Next, reads are aligned *de novo* or to a reference genome and nucleotides for each individual are counted at locations with significant read depth. These counts are used in a maximum likelihood framework to assign genotypes and, consequently, identify SNPs (Baird et al. 2008; Etter et al. 2011; Catchen et al. 2011; Catchen et al. 2013). This method provides the following advantages: it does not require any prior genomic information for the taxa being studied, it can sequence higher numbers of samples cost effectively, and it has a greater depth of coverage per locus, improving confidence in genotype calls (Andrews et al. 2016).

Individual genotype assignment and SNP discovery has become a powerful tool that is used to address a variety of scientific questions. SNPs have been applied to the medical field as a way of implementing precision medicine (Rocha et al. 2020; Adolf et al. 2021). In ecology, SNPs have been used to infer population structure and genetic diversity, comparative genetic diversity, gene selection, adaptation, effects of landscape variables on gene flow, and conservation and management strategies in a broad range of taxa including cattle, polar bears, bobcats, Atlantic mackerel, brown trout, yellow perch, and walleye (Rodríguez-Ezpeleta et al. 2016; Makina et al. 2014; Stepien et al. 2015; Storfer et al. 2018; Lemopoulos et al. 2018; Kozakiewicz and Grzybowska-Szatkowska 2018; Viengkone 2015).

The present study aims to investigate the genetic background of walleye sampled from Iowa hatchery broodstocks. Specifically, walleye originating from lake environments were compared with walleye originating from river environments. Findings will be used to evaluate each strain in large reservoir environments and, ultimately, to

offer stocking guidance to fisheries management. As previously stated, walleye stocking dates to the 1950's and includes numerous stocks into Iowa waterbodies; therefore, a powerful tool is necessary for delineating the genetic complexity. SNP data is suitable for strongly and weakly diverged populations, as evidenced by salmonids (Estoup et al. 1998; Vähä et al. 2007) and can reveal ancestral patterns of genetic structure because of a slower mutation rate (when compared to microsatellites). Because the specimens sampled were not wild, and instead are from a highly managed facility, populations under investigation are referred to as "groups" with subpopulations named "subgroups".

Study System and Sample Collection

Fish sampling was conducted at Rathbun Fish Hatchery in Moravia, IA during the spring 2019 spawning season. Pelvic fin tissue was collected from known lake-strain and river-strain adult walleye (*Sander vitreus*) and stored in 95%+ ethanol. The samples in this dataset comprised a total of 69 specimens, 33 lake-strain individuals and 36 river-strain individuals. Within the lake-strain group there are four subgroups (Table 1): Old Lake Strain (OLS) including 15 individuals from year classes 2008, 2009, and 2015, Lake Strain Outside (LS17_Out) including 6 individuals hatched in 2017 and raised in an outside facility, Lake Strain RAS (LS17_RAS) including 6 individuals from the year class 2017 and raised in a recirculating aquaculture system (RAS), and Lake Strain 2018 (LS18) including 6 individuals from the year class 2018. Within the river-strain group there are three subgroups (Table 1): Female River Strain 2017 (RS17_F) and Male River Strain 2017 (RS17_M) each including 15 individuals from the year class 2017, and River Strain 2018 (RS18) including 6 individuals from the year class 2018.

Library Preparation and Sequencing

The RAD-seq protocol developed by Etter et al. (2011) and modified by Gamble et al. (2015) and Luiken et al. (2021) was used to prepare RAD-seq libraries for sequencing. In brief, DNA was extracted, quantified, and digested with high-fidelity *SbfI* restriction enzyme (New England Biolabs). A two-adapter process was used to create RAD-seq libraries to provide unique Illumina identifiers. P1 adapters were given barcodes, or multiplex identifiers, specific to each sample and then ligated onto the *SbfI* cut site. Samples were pooled, with 15 individuals per library, sheared, and size-selected to 400-600 bp fragments using a Pippin Prep size selection system (Sage Science, Beverly, MA).

The resulting fragments were blunt-end repaired and dA-tailed to prepare the libraries for ligation to the P2 adapter, which contained unique Illumina barcodes. Genomic libraries were amplified via PCR, cleaned, and size-selected again using a GeneRead Size Selection Kit (Qiagen). The libraries were quantified and pooled to prepare for sequencing. DNA fragment sizes were verified using a Bioanalyzer (Agilent, Santa Clara, CA). Pooled libraries were sequenced on NovaSeq 6000 (Illumina, San Diego, CA) at the Roy J. Carver Center for Genomics, University of Iowa using 100 bp paired-end reads.

RAD Data Assembly and SNP Discovery

Loci were identified and genotyped using *STACKS v.2.54* following the protocols of Catchen et al. (2013). The *process_radtags* script was used to demultiplex and quality filter raw Illumina reads. Reads were truncated to a length of 83 bp, *'-t 83',* to trim low-

quality bases at the 5′-end, remove P2 adapter sequence, and ensure reads were the same length. A mismatch of 2 was allowed in the adapter sequence, *'--adapter_mm 2'*.

The STACKS pipeline for non-reference aligned data, *denovo_map.pl,* was used to build loci because a fully annotated and scaffolded genome was not yet available *S. vitreus*. This identified SNPs from stacks of sequence reads and created a catalog of loci across all samples according to the population map, a text file assigning individuals to populations, or groups. Parameters were optimized through 5 trials to maximize number of SNPs with quality coverage and to minimize rates of error using '*-m 4',* the minimum number of identical raw reads required to create a stack, *'-M 3',* the maximum number of mismatches allowed between loci when processing a single individual, and *'-n 2',* the maximum number of mismatches allowed between loci when building the catalog (Catchen et al. 2013; Paris et al. 2017).

Dataset Generation

The *populations* script from *STACKS v.2.54* was run to analyze each group of individuals, computing population genetics statistics and creating output files for downstream analyses. Instead of filtering for missing data, STACKS has implemented the following flags to filter for present-ness: *'-r 0.80'* and *'-p 1'*. This ensured only SNP loci shared by a minimum of 80% of all individuals were retained, and that a locus had to be present in at least one group to be processed. The parameters *'--min-maf 0.05'*, *'--max-obs-het 0.50'*, and *'--write-single-snp',* filtered alleles occurring at a frequency of <5%, filtered artefactual nucleotide sites at a locus that had an observed heterozygosity >50% due to paralogous loci or multilocus contigs, and restricted data analysis to only the first

SNP per locus to minimize potentials for linkage disequilibrium (Catchen et al. 2011; Catchen et al. 2013; Gargiulo et al. 2020). Additional flags were executed to produce formats suitable for downstream bioinformatic analyses.

Following the assembly of the initial dataset, three SNP datasets were constructed using the parameters outlined above: one for both strains combined, one to analyze lake-strain broodstock independently, and one to analyze river-strain broodstock independently. Hereinafter, the datasets will be referred to as the LR dataset, which contains lake- and river-strain individuals and all their subgroups; the lake dataset, which contains lake-strain individuals belonging to subgroups OLS, LS17_Out, LS17_RAS, and LS18; and the river dataset, which includes river-strain individuals belonging to subgroups RS17_F, RS17_M, and RS18.

CHAPTER 3

GENETIC DIVERSITY OF BROODSTOCK

Introduction

As stocking strategies continue to be optimized, genetic diversity, structure, and strain performance remain important limiting factors (Paragamian 1988). Populations that are supported by stocks and harvested from a combined fishery are especially hard to preserve diversity in (Vinther et al. 2004; Chen et al. 2020). Such is the case for walleye, as shown by the example in Great Lakes fisheries (Kocovsky et al. 2013; Stepien et al. 2015). Chen et al. (2020) used RAD-Seq to delineate population structure of walleye in Lake Erie. They were unable to differentiate within-basin genetic structure, likely due to sufficient straying of fish during spawning seasons but were able to classify samples to a lake basin with 95% accuracy.

The genetic effects of walleye management in Iowa have been investigated by few groups. A study by Paragamian (1988) concluded that despite 35 years of stocking walleye from Spirit Lake, IA into the Cedar River, fish from the Cedar and Mississippi rivers were genetically similar and were significantly different from walleye in Spirit Lake. Mississippi River-strain walleye have been noted to have a more substantial impact on survival and recruitment when stocked into Iowa interior rivers than Spirit Lake-strain individuals (Gelwicks 2001; Gelwicks et al. 2019). Due to Iowa's complex stocking history, comparable results are hypothesized to be seen in the present study.

Methods for Diversity Analysis

Diversity statistics were performed on the combined LR dataset, the lake-strain only dataset (L dataset), and the river-strain only dataset (R dataset). Subgroups of each dataset were considered when calculating the diversity indices. Population heterozygosity, mean allele frequency, nucleotide diversity, and F statistics were obtained from the STACKS *populations* pipeline similar to Whelan et al. (2019) and Schmidt et al. (2021). The analysis of molecular variance F_{ST} calculation was derived from Weir (1997). A Fisher's Exact Test (FET) p-value was recorded to describe if the F_{ST} measure was statistically significant (Catchen et al. 2013). A p-value was calculated for each variable position between pairs of populations to test statistical significance. FET was preferred because it is calculated using the exact null randomization distribution, the procedure which led to the observed data, and can differ from the Student's *t* distribution (Bind and Rubin 2017).

Genetic structure was assessed for each dataset using three different methodologies: discriminant analysis of principal components (DAPC), Bayesian assignment test, and genetic relationship matrices (GRM). The discriminant analysis achieves the best discrimination of individuals into pre-defined groups by yielding synthetic variables, or the structure retained in the principal components, which maximize the between-group variation. This allows for a probabilistic assignment of individuals into pre-defined groups, like in Bayesian clustering algorithms, but performs generally better and quicker than STRUCTURE at characterizing population subdivision (Jombart et al. 2010). The DAPC was conducted with the R package *adegenet* to identify

and describe genetically similar individuals and identify genetic clusters (Jombart 2008; Jombart et al. 2010; Jombart and Collins 2015). The DAPC transforms multilocus genotype data by using principal component analysis (PCA) to derive the uncorrelated variables that serve as input for discriminant analysis (DA). This method deals with noise well by standardizing the data and using sequential K-means and model selection to infer genetic groups (Viengkone 2015). After determining the optimal number of clusters for each dataset, cross-validation was applied to ensure the data was not overfitted.

Detection and visualization of genetic structure was produced with a Bayesian assignment test via fastSTRUCTURE that works on population data that is not defined *a priori*. fastSTRUCTURE, a variant of STRUCTURE, was developed to efficiently infer population structure of large SNP datasets by using variational Bayesian framework algorithms and has been shown to have the same accuracy (Raj et al. 2014). The *structure.py* command was run at K values ranging from 1-8 on the LR dataset. The *'-cv 500'* flag was used to run 500 test sets for cross validation at each K value. The same parameters were used when running the command for the lake and river datasets, except K values ranged from 1-4 and 1-3 for each subgroup, respectively. The *chooseK.py* command was implemented to choose the appropriate number of model components that best explained the structure in each of the datasets (Raj et al. 2014). The *distruct.py* command was executed to visualize the expected admixture proportions inferred for all three datasets.

Plink v1.9 (Chang et al. 2015) was utilized to create a GRM for each dataset. The '*distance square identity by state*' (ibs) command generates a pairwise comparison of

allele counts that was visualized with an interactive heatmap on the Morpheus matrix analysis platform (Morpheus, https://software.broadinstitute.org/morpheus/). The color ramp correlates to the minimum and maximum value of each row independently, presenting a relative color scheme. The heatmap allowed for detection of patterns of relatedness, or un-relatedness, in the data.

Results

Approximately 140 million 100bp paired-end reads were generated. The final datasets included 11,764 variant sites that passed all quality filters in the LR dataset, 10,243 variant sites in the Lake dataset, and 6,598 variant sites in the River dataset. All 69 individuals had a depth of coverage ranging from 13.5x - 28x after filtering, therefore all 69 individuals were used in the analyses.

Descriptive Statistics Summary

Descriptive statistics quantified the genetic diversity in the LR, Lake, and River datasets. Within the combined LR dataset the genetic diversity was similar among year classes within each strain. However, there were differences observed between strains (Table 1A). Overall, the river-strain had a higher genetic diversity (H_o=0.262 avg.) than lake-strain (H_o=0.243 avg.), but the difference was minimal. Within lake-strain subgroups OLS had the most genetic diversity (H_o=0.260), which was expected due to multiple year classes making up the subgroup. The lowest diversity belonged to LS17_Out (H_o=0.230), which was different from other individuals from the same year class LS17_RAS (H_o=0.243). Within river-strain subgroups, the males had the most genetic diversity (H_o=0.301), which was higher than the females from the same year class, RS17_F

(H_o=0.282). Subgroup RS18 had the lowest genetic diversity (H_o=0.275), but the difference was minimal. The F_{ST} comparisons revealed no significant divergence among strains or subgroups in all three datasets. Within the LR dataset, the largest differentiation was found between subgroup LS17_Out and the river-strain year class 2017 subgroups (Table 2A). Subgroups RS17_F and RS17_M were the most similar.

Table 1. Genetic diversity characteristics of Iowa walleye broodstock collected from Rathbun Hatchery, IA in 2019. A. Combined LR dataset. B. Lake dataset. C. River dataset. Estimates included number of samples (N), number of alleles analyzed (N_a), observed heterozygosity (H_O), expected heterozygosity (H_E), mean allele frequency (P), nucleotide diversity (π), and inbreeding coefficient (F_{IS}).

A. Combined LR dataset

Spawning Stock	Year Class	Stock Source	N	N_a	H_O	H_E	P	π	F_{IS}
OLS	2008, 2009, 2015	Rathbun, Iowa	15	14,930	0.260	0.263	0.817	0.273	0.044
LS17_Out	2017	Rathbun, Iowa	6	10,253	0.230	0.236	0.831	0.259	0.065
LS17_RAS	2017	Rathbun, Iowa	6	14,266	0.243	0.241	0.831	0.265	0.049
LS18	2018	Rathbun, Iowa	6	11,255	0.240	0.233	0.835	0.255	0.034
RS17_F	2017	Genoa, Wisconsin	15	11,021	0.282	0.240	0.831	0.249	-0.077
RS17_M	2017	Genoa, Wisconsin	15	9,992	0.301	0.248	0.824	0.257	-0.095
RS18	2018	Genoa, Wisconsin	6	12,349	0.275	0.254	0.818	0.280	0.010

B. Lake dataset

Spawning Stock	Year Class	Stock Source	N	N_a	H_O	H_E	P	π	F_{IS}
OLS	2008, 2009, 2015	Rathbun, Iowa	15	15,127	0.277	0.285	0.804	0.296	0.057
LS17_Out	2017	Rathbun, Iowa	6	10,517	0.250	0.261	0.815	0.286	0.081
LS17_RAS	2017	Rathbun, Iowa	6	14,508	0.260	0.262	0.818	0.287	0.061
LS18	2018	Rathbun, Iowa	6	11,482	0.260	0.256	0.820	0.281	0.045

C. River dataset

Spawning Stock	**Year Class**	**Stock Source**	**N**	**N_a**	**H_O**	**H_E**	**P**	**π**	**F_{IS}**
RS17_F	2017	Genoa, Wisconsin	15	8,575	0.285	0.258	0.826	0.268	-0.043
RS17_M	2017	Genoa, Wisconsin	15	7,480	0.302	0.263	0.821	0.273	-0.062
RS18	2018	Genoa, Wisconsin	6	10,271	0.286	0.275	0.805	0.303	0.038

Due to the parameters set when generating each dataset, specifically *'-r 0.80'*, more resolution was found when analyzing lake and river datasets independently. The Lake dataset had lower heterozygosity overall when compared to the River dataset (Table 1B and 1C). Within the Lake dataset, the genetic diversity was similar among subgroups. LS17_Out had the lowest genetic diversity despite having moderate nucleotide diversity, compared to other subgroups. Subgroup OLS appeared to have the most genetic and nucleotide variation (H_o=0.2769; π=0.2958). The inbreeding coefficient indicated a deficit in heterozygosity among lake-strain subgroups. In the lake dataset, F_{ST} values ranged from 0.0281-0.0542 (Table 2B). The most closely related subgroups were OLS and LS17_Out while the comparison between LS17_Out and LS18 revealed the greatest amount of variation.

Table 2. Comparison of F_{ST} values by subgroup. A. Combined LR dataset. B. Lake dataset. C. River dataset.

A. Combined LR dataset

	OLS	LS17_Out	LS17_RAS	LS18	RS17_F	RS17_M	RS18
OLS	-						
LS17_Out	0.0285	-					
LS17_RAS	0.0276	0.0475	-				
LS18	0.0285	0.0534	0.0474	-			
RS17_F	0.0863	0.0935	0.0883	0.0882	-		
RS17_M	0.0857	0.0915	0.0869	0.0856	0.0159	-	
RS18	0.0565	0.0854	0.0808	0.0782	0.0828	0.0780	-

B. Lake dataset

	OLS	**LS17_Out**	**LS17_RAS**	**LS18**
OLS	-			
LS17_Out	0.0297	-		
LS17_RAS	0.0281	0.0478	-	
LS18	0.0293	0.0542	0.0476	-

C. River dataset

	RS17_F	**RS17_M**	**RS18**
RS17_F	-		
RS17_M	0.0171	-	
RS18	0.0896	0.0858	-

Within the River dataset, the overall average observed heterozygosity and mean allele frequency were similar across the subgroups (Table 1C). Subgroup RS17_M had the highest observed genetic diversity (H_O=0.301); however, subgroup RS18 had the highest nucleotide diversity (π=0.3029). The inbreeding coefficient indicated a deficit in heterozygosity in the RS18 subgroup and an excess in RS17_M and _F subgroups. The greatest amount of divergence was observed between the RS17 and RS18 subgroups (Table 2C).

Genetic Structure Results

The DAPC of the LR dataset revealed a complex population structure. Bayesian Information Criterion (BIC) scores were lowest for K = 3 using 25 principal components and four discriminant functions, revealing 3 distinct genetic groups. These parameters proved optimal upon cross validation as the combination had the lowest root mean square error (RMSE) among trials. Discriminant functions 1 and 2 were used for visualization because they explained the most variation between clusters (Figure 2A). This representation of between-group structure was able to differentiate between the lake- and river-strains. Within the river-strain, RS17 and RS18 formed distinct genetic clusters and contained genetic variation within each cluster. Within the lake-strain, all the subgroups were completely overlapping indicating a single genetic cluster. LS18 exhibited the most variation within the lake-strain subgroups (Figure 2A).

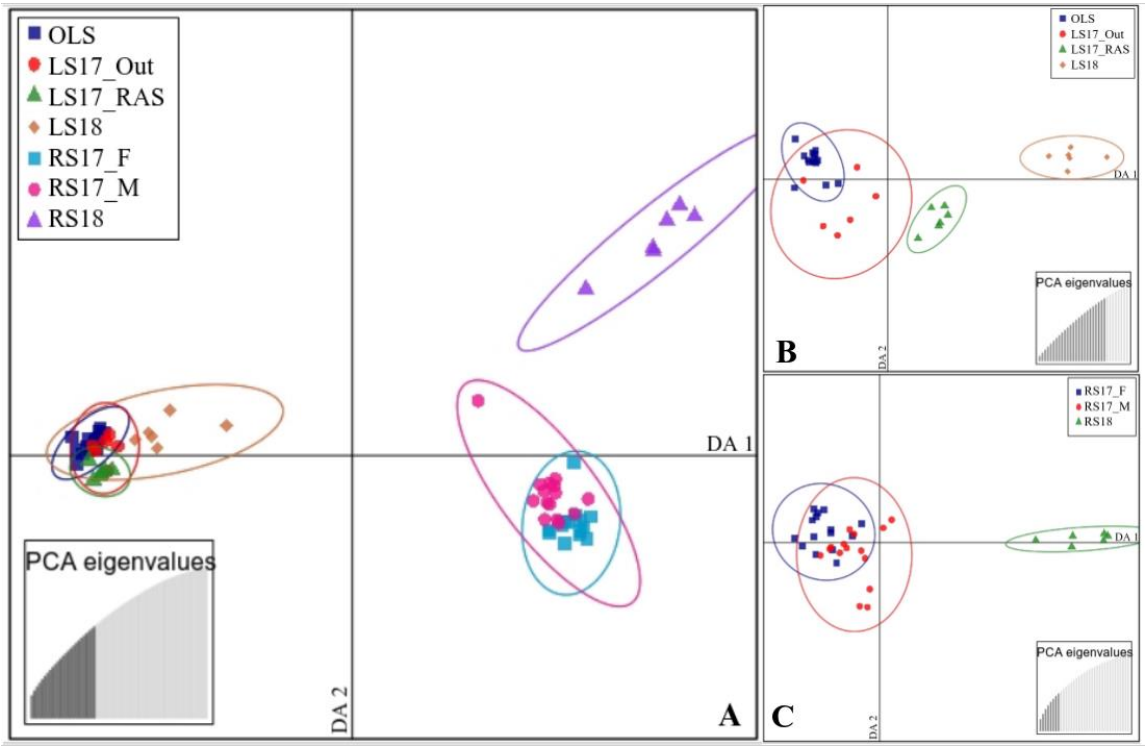

Figure 2. Scatterplots from discriminant analysis of principal components (DAPC) with individuals designated by locality. The first two discriminant analyses (DA) are plotted for each dataset. A. LR dataset based on the first 25 principal components with K=3. B. Lake dataset based on the first 24 principal components with K=3. C. River dataset based on the first 8 principal components with K=2.

The DAPC of the Lake dataset revealed 3 distinct genetic groups (Figure 2B). BIC scores were lowest for K = 3 using 24 principal components and 3 discriminant functions. The greatest variation within subgroups was found in LS17_Out, which overlapped with subgroup OLS. Both OLS and LS17_Out were more closely related with LS18. Yet, subgroups LS17_RAS and LS18 formed distinct subgroups. Within the River dataset the DAPC identified 2 genetic clusters (Figure 2C). BIC scores were lowest for K = 2 using 8 principal components and 2 discriminant functions. The subgroups RS17_M

and _F formed a cluster that was distinct from RS18. All river-strain subgroups had high within-group variation.

The Bayesian assignment test indicated distinct genetic clusters in all three datasets. In the LR dataset, 3 genetic clusters were identified that maximized marginal likelihood with 2 model components used to explain the structure (Figure 3). When comparing 2 and 3 genetic clusters to describe the dataset, it was found that the third genetic cluster came from a single individual in the RS18 subgroup. Because a Bayesian assignment test is optimal for finding global patterns of structure and is aggressive in removing model components that seem unnecessary, K=3 was used to explain the genetic clusters in the LR dataset (Raj et al. 2014). All lake-strain individuals contained one genetic cluster. Only river-strain individuals contained admixture between clusters, one shared with lake-strain individuals and two that were distinct. The third cluster, shown in orange, contains one individual from the RS18 subgroup (Figure 3). All samples passed conservative filtering and contained credible depth of coverage to be considered for analyses. Therefore, the third cluster was kept in the analysis because it contained credible genetic variation that was distinct from other individuals.

The LR dataset GRM is presented by the heatmap in Figure 3. Between the two strains there is a moderate to high degree of relatedness. Within the lake-strain there is a high degree of relatedness, which is consistent across all individuals. However, within the river-strain there is greater variability in relatedness. Within RS17 there is a high degree of relatedness except for a few individuals (RS009, RS073, RS076). Individuals within

RS18 show minimal relatedness with the other river-strain individuals and the lake-strain individuals.

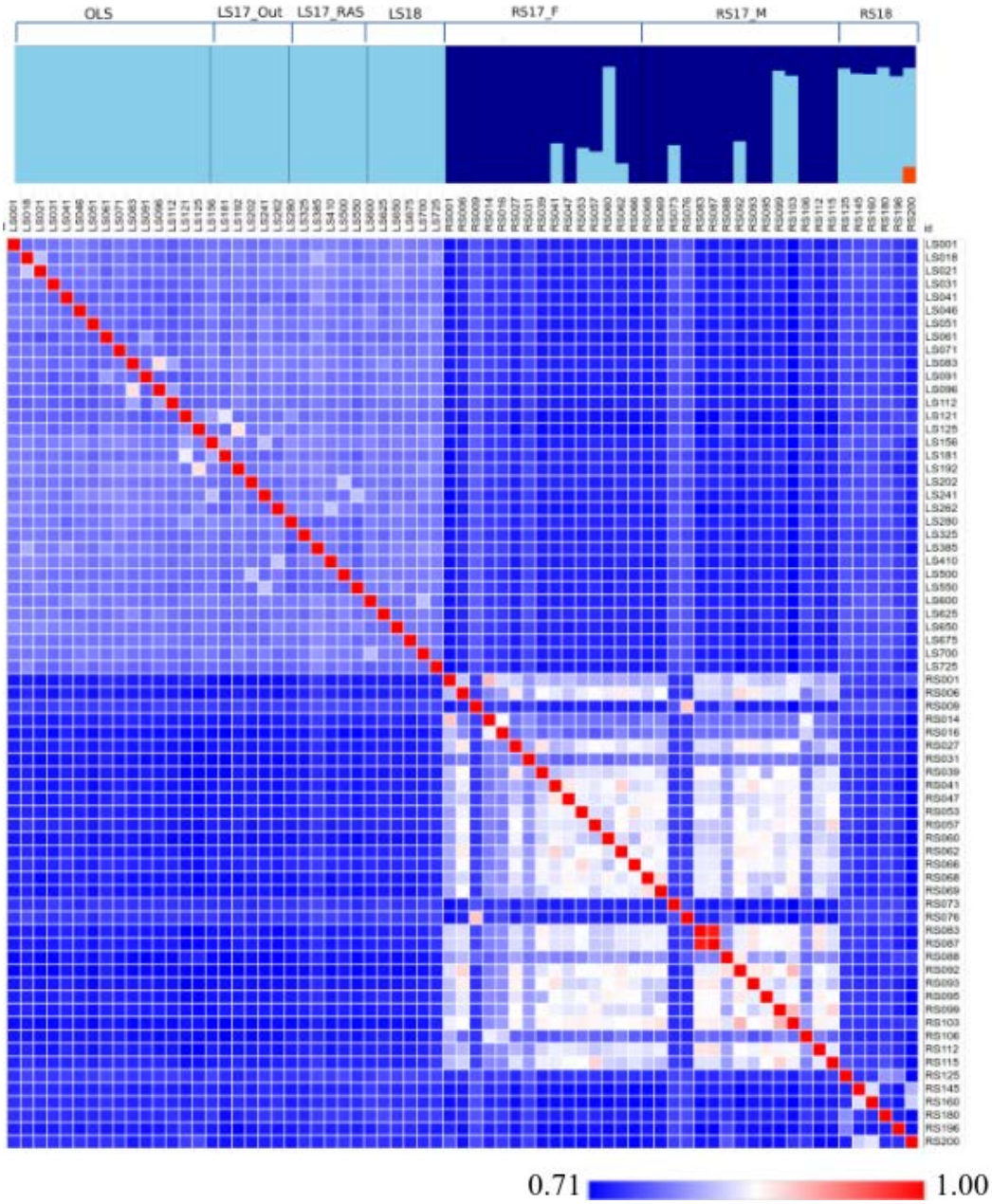

Figure 3. Genetic structure of the LR dataset. Above, Bayesian assignment test with k-means clustering (K=3). The x-axis represents individuals sorted according to sample ID within subgroups. The colors represent admixture from 3 genetic clusters (light blue, dark blue, orange). On the y-axis each individual is represented by a vertical line partitioned into colored segments with lengths proportional to admixture from ancestral populations. Below, a heatmap of the GRM. Individuals are along the x-axis of the Bayesian Assignment test and correspond moving down the heatmap. The color gradient of the heatmap represents the genomic relationship between pairs of individuals with values mapped to colors using the minimum (blue) and maximum (red) of each row. The minimum and maximum value of the dataset is presented on the color bar legend.

In the Lake dataset, the Bayesian assignment test identified that 1 genetic cluster maximized marginal likelihood with 2 model components used to explain the data. Because the DAPC revealed multiple clusters within the lake dataset (Figure 2B), K=2 was chosen because it was known that there was more than 1 genetic ancestry present in the data (Figure 4). Although both genetic clusters were found within each subgroup (dark and light blue), admixture within individuals is not present given the data. When comparing this result with the findings of the DAPC, the DAPC depicted finer-scale clustering including three distinct groupings. The GRM showed no distinct clustering within the lake dataset, with all individuals containing a similarly moderate to high level of relatedness.

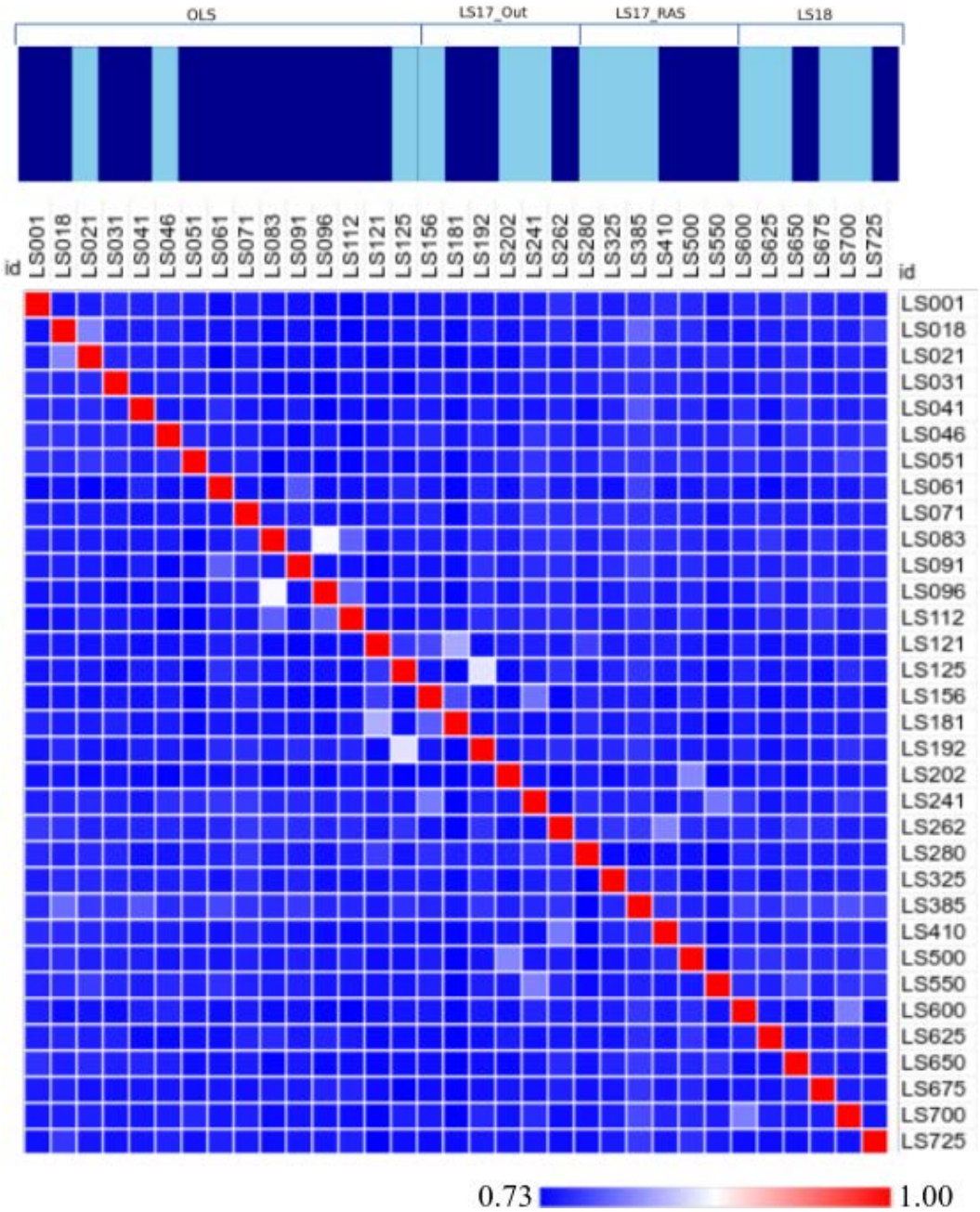

Figure 4. Genetic structure of the Lake dataset. Above, Bayesian assignment test with k-means clustering (K=2). The x-axis represents individuals sorted according to sample ID within subgroups. The colors represent admixture from 2 genetic clusters (light blue and dark blue). On the y-axis each individual is represented by a vertical line partitioned into colored segments with lengths proportional to admixture from ancestral populations. Below, a heatmap of the GRM. Individuals are along the x-axis of the Bayesian Assignment test and correspond moving down the heatmap. The color gradient of the heatmap represents the genomic relationship between pairs of individuals with values mapped to colors using the minimum (blue) and maximum (red) of each row. The minimum and maximum value of the dataset is presented on the color bar legend.

In the River dataset, the Bayesian assignment test identified 2 genetic clusters maximized marginal likelihood with 3 model components used to explain the data. Based on the variation between and within clusters revealed by the DAPC (Figure 2C), K=3 was chosen because it revealed strain-specific genetic ancestries (Figure 5). Individuals from the year class 2017 contain admixture from two genetic clusters (dark and light blue) while individuals from year class 2018 contain admixture from a shared cluster (dark blue) and from a distinct genetic ancestry (orange). The detection of variable genetics in 2018 fish also supports the choice to keep the RS200 individual as the third cluster found in the LR dataset. The GRM revealed distinct clusters of related individuals within subgroups, which agrees with the clustering in the DAPC.

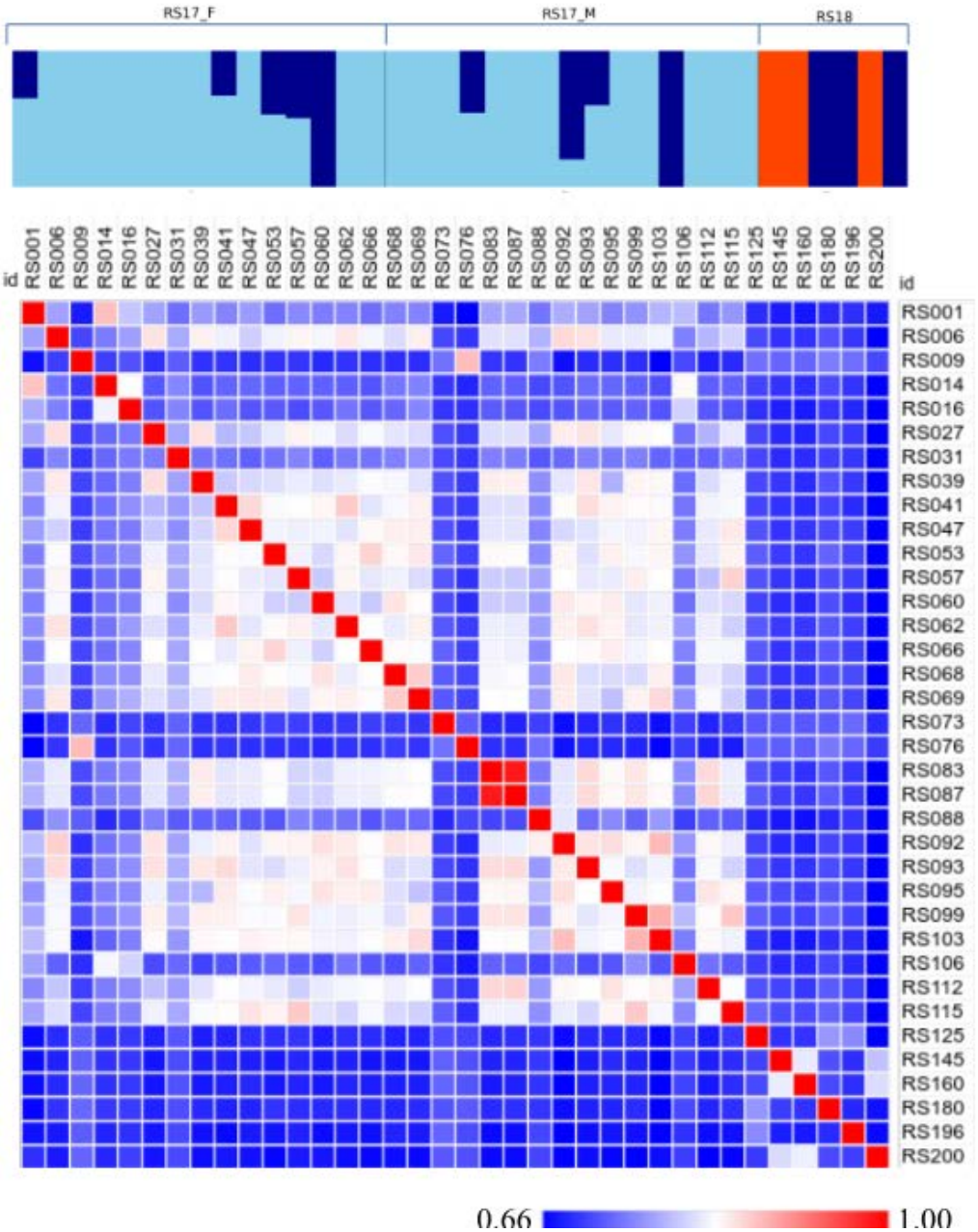

Figure 5. Genetic structure of the River dataset. Above, Bayesian assignment test with k-means clustering (K=3). The x-axis represents individuals sorted according to sample ID within subgroups. The colors represent admixture from 3 genetic clusters (light blue, dark blue, orange). On the y-axis each individual is represented by a vertical line partitioned into colored segments with lengths proportional to admixture from ancestral populations. Below, a heatmap of the GRM. Individuals are along the x-axis of the Bayesian Assignment test and correspond moving down the heatmap. The color gradient of the heatmap represents the genomic relationship between pairs of individuals with values mapped to colors using the minimum (blue) and maximum (red) of each row. The minimum and maximum value of the dataset is presented on the color bar legend.

Discussion

The goal of this study was to determine the genetic diversity of Iowa walleye broodstock. Genome-wide SNP datasets demonstrated the ability to detect fine-scale structure both between and within genetic strain. The results of the analyses support a genetic differentiation between the two Iowa walleye strains - lake and river; however, this variation is subtle. Analyses of the lake-only dataset revealed that lake-strain individuals within different subgroups share similar ancestry while analyses of the river-only dataset revealed that individuals contained genetic variation between year class individuals.

Combined LR Dataset

Analyses of the combined LR dataset revealed that the two strains were genetically distinct, evidenced by the DAPC (Figure 2A) and Bayesian assignment test (Figure 3). The DAPC displayed no evidence of shared clusters between strains, meaning individuals had a higher grouping power within strain than between. The Bayesian assignment test revealed that the river-strain contained ancestral components that were not shared by the lake-strain, which was further supported by the consistently lower relatedness values when lake- and river-strain individuals were compared (Figure 3).

Genetic differentiation between strains was expected due to the history of Iowa stocking. Lake stocking events dating back to the 1880's included fish sourced from both Spirit Lake and the Mississippi River. However, only stocking of lake-origin fish continued with the implementation of the lake-strain in Rathbun Reservoir, allowing a minor addition of different genes to the many generations of walleye in this closed

system (Paragamian 1988). Lake-strain walleye have had many years of population management, but the river-strain has only been implemented and maintained within the last decade. The results show that river-strain fish have higher variability, which may be due to their ancestral habitat range in the Mississippi River, allowing access to multiple wild stocks. Likewise, the lower variability observed in lake-strain walleye could be due to a smaller habitat range within Iowa lakes, decreasing the amount of migration and potential for gene flow.

Although it was found that the lake- and river-strains were genetically different, the degree of differentiation was low. Many individuals contained moderate to high levels of relatedness, which were supported with similar values of descriptive statistics among subgroups. Specifically, all individuals shared 71% or more of alleles, denoted by the minimum relatedness score (Figures 3). No F_{ST} values were statistically significant, meaning allele counts between subgroups of each strain were more likely to be the same than different. The result of strain-distinct genetic groups, yet no statistically significant differences is likely due to the variation within strain. Analyses of more individuals would give a better understanding of variation present between and within strains, which could potentially identify significant differences.

Both strains contained variability between and within subgroups with river-strain subgroups containing more variation than lake-strain subgroups (Table 1A). These results were similar to what was observed in walleye from Lake Erie stocks (Euclide et al. 2021). Within the river-strain, each year class formed a distinct cluster, revealing unique ancestries in river fish from 2017 and 2018. The Bayesian assignment test revealed that

one individual, RS200, from the RS18 subgroup contained an additional genetic signature. Inspection of the heatmap and the DAPC confirm that there is unique ancestry within the RS18 subgroup.

River-strain broodstock individuals typically come from a variety of sources along the Mississippi River (Gelwicks et al. 2019). The individuals sampled in the present study have an origin of the Mississippi River near Genoa, WI. Eggs fertilized by wild males were sent to Rathbun State Fish Hatchery from the Genoa National Fish Hatchery in 2017 and 2018. The number of eggs shipped in 2018 was twice that from 2017 (Dagendesh and Aloisi 2017; Dagendesh and Aloisi 2018). Eggs obtained from spawning individuals occurred at the beginning of the spawning event in 2017 and during the peak of the spawning event in 2018. These factors could explain the added variability in the RS18 subgroup due to the higher number of wild females that added to the egg supply and the likeliness of sampling different individuals between years.

Lake Dataset

Analyses of the lake-only dataset consistently revealed that individuals are contained in a single genetic cluster, indicating common ancestry, and that they have lower variability than individuals of the river-only dataset (Table 1). Within this dataset, two genetic clusters were identified (Figure 4); however, no genetic cluster was identified contained within a specific subgroup.

The DAPC was able to detect finer-scale resolution (Figure 2B). Although the lake-strain subgroups share genetic ancestries, the DAPC was able to delineate three genetic clusters. Based on this analysis, subgroup LS17_RAS was the most genetically

distinct. Despite containing individuals of the same year class, subgroup LS17_Out was more closely related to subgroup OLS than subgroup LS17_RAS. However, these differences were subtle as shown by low to moderate F_{ST} values with no statistical significance (Table 2). The history of lake-strain stocking supports the finding of shared ancestry among all subgroups, due to broodstock collected from a variety of Iowa lakes (Gelwicks et al. 2019).

River Dataset

Analysis of two year classes (2017 and 2018) and sex within year class 2017 was included in the analysis of the river-only dataset. Year class 2017 was consistently found to be genetically distinct from year class 2018 by the DAPC (Figure 2A and 2C), Bayesian assignment test, and GRM (Figure 5). The Bayesian assignment analysis revealed that one genetic cluster was contained in all subgroups (dark blue) and one genetic cluster was distinct to each year class (shown by light blue in 2017 and orange in 2018). The distinct clusters could be explained by the previously mentioned sampling differences during spawning events in 2017 and 2018. Due to the comparatively higher F_{ST} values between year class subgroups, it was concluded that the strain-specific genetic clusters heavily impact the genetic variation between the year class subgroups. The average F_{ST} value between the river-strain year classes 2017 and 2018 (0.0877) was larger than the average F_{ST} value between river- and lake-strain individuals (0.0839), revealing that given the data there is greater variation between river-strain year classes than between lake-strain and river-strain, although this difference is minimal.

Sex was also examined across the river-strain year class 2017 for genetic variation. The DAPC revealed shared genetic clustering and high relatedness, which was supported by a low F_{ST} between subgroups RS17_F and RS17_M (Figure 2C and Table 2C). The Bayesian assignment test was unable to distinguish males and females (Figure 5). The GRM supported that there was moderate to high relatedness values between male and female individuals, except for 3 individuals with lower values. Biologically, this makes sense as most fish sampling takes place during spawning periods when males and females are in the same areas. Walleye are known to show site fidelity during the spawning period, therefore, allowing for the admixture of genes from the same adults, generation after generation (Chen et al. 2020; Hayden et al. 2017).

Conclusion

The results of this study demonstrate the power of genome-wide SNPs and their ability to distinguish between weakly-diverged populations, or subgroups. Most research on walleye broodstocks has been done using a 'top-down' format where wild spawning stocks with unknown population identity and structure are sampled and investigated. However, this study presents a 'bottom-up' approach where individuals' groups are known due to sampling from managed broodstocks. Knowing the genetic signature of the broodstocks prior to stocking allows for a better understanding of individuals that are surviving in the wild.

Waltner (1988) investigated the genetics of a South Dakota walleye broodstock, comparing them with upper Mississippi and Missouri river stocks, and found genetic dissimilarity between South Dakota stocks and those of the Mississippi River. From this,

they cautioned against promiscuity in stocking of mixed strains to prevent the loss of distinctive characteristics. The study referenced by Chen et al. (2020) was performed on Lake Erie walleye and concluded that within western basin stocks contained low genetic divergence while comparisons between the western and easter basin stocks revealed more genetic structure (F_{ST}=0.0042-0.0064). Euclide et al. (2021) used genomic data to analyze harvest of Lake Erie's commercial and recreational walleye. Despite the weakly-diverged stocks (F_{ST}=0.0018-0.0342), they traced the origin of harvested fish revealing that western basin stocks were harvested more during peak walleye fishing season and eastern basin stocks were harvested earlier in the season. These studies showed similar genetic variation to our results and their findings provided great insight into estimated stock contributions.

Stocking without the realization of genetic differentiation, and the relationship between this differentiation and fitness, may result in cross breeding that limits, or eliminates, genetic diversity (Murphy and Lee 1986). Genetic variation creates a buffer against negative pressures by keeping a "portfolio" of population diversity (Schindler et al. 2010; DuFour et al. 2015; Chen et al. 2020). Therefore, to promote plasticity and survivability, discrete stocks should be identified and their unique traits recognized, especially when used as broodstock (Waltner 1988). This may be evident in the present example as inbreeding shows a deficit in heterozygotes in the heavily-managed subgroups (F_{IS} = 0.0336-0.0649) and an excess in heterozygotes in the river-strain subgroups (F_{IS} = -0.0946-0.0102). Heterozygosity is an important indicator of genetic variation within populations. Further research is needed to evaluate whether this human-

induced variability is aiding or inhibiting the genetic diversity of Iowa walleye broodstock.

CHAPTER 4

SIGNIFICANT MARKER IDENTIFICATION

Introduction

A proper identification mark is retained, easily identifiable, and does not affect capturability or survivability of the individual (Stott 1968; Wydoski and Emery 1983; Sandford et al. 2020). However, traditional identification methods often do not meet these criteria and instead present many drawbacks. Any procedure that involves handling of the individual impacts the organism's welfare; therefore, managers aim for minimal fish handling. External and internal tags, freeze branding, and fin clipping are examples of common practices used today. Drawbacks of these methods include uncertain impacts on growth and survival, infection due to exposure to opportunistic bacterium, lost tags resulting in money and data loss, and misidentification due to unidentifiable marks (Sandford et al. 2020; Uglem et al. 2020; Axelsson et al. 2020). Each procedure involves varying time devotion due to both the handling of individual fish at initial marking and the individual evaluation required for identification upon recapture.

Genetic markers for stock identification have been used in the past (Uthe et al. 1966; Billington 1996; Glover et al. 2010) and have improved due to advancements in the application of next generation sequencing methods. RAD-Seq generates large SNP datasets that represent the whole genome. RADSex, a computational workflow, was designed to study and visualize the genetic basis of sex determination using RAD-Seq data. It first finds sex-specific genomic sequences, aligns them to a reference genome if available, finds the locus of interest, and then can identify candidate master sex-

determining (MSD) genes, among other applications (Feron et al. 2021). Although it was created to identify sex-specific markers, it was deliberately designed for the possibility of use on a broader scale. No fish-specific assumptions were used in the implementation of RADSex, allowing for application on a variety of taxa. The generic workflow can be applied to any major quantitative trait loci with contrasting binary phenotypes.

When comparing binary traits, SNPs that are significantly associated with one trait and not the other can be used for differentiation, and subsequently, identification. This study used the RADSex method to distinguish between two phenotypes, strain and sex.

Methods

Concatenated output files from the primary filtering step of the STACKS (Catchen et al. 2013) pipeline were utilized in the RADSex analysis. Two binary traits were investigated - strain and sex - through three different analyses: comparison of lake-strain and river-strain "phenotypes" in all individuals (n=69), comparison of lake-strain and river-strain phenotypes in all individuals excluding those from the RS18 subgroup (n=63), and comparison of sex within the RS17 subgroups (n=30). The analysis that involved the removal of the RS18 individuals, n=63, was considered due to the results of the genetic diversity analyses. Because river-strain individuals clustered into two groups, compared to the one cluster of lake-strain individuals, it was determined that removing the unique river-strain individuals may yield more significantly different variants between the lake- and river-strain phenotypes.

For the analysis of the lake-strain or river-strain phenotype in all individuals, a table of marker depths was computed via the RADSex *process* command using default

parameters. Unique sequences found throughout the entire dataset were organized into a table with an identifier and the depth at which the sequence was found in each individual. RADSex organizes only monomorphic markers because alleles at polymorphic loci are treated as separate markers. The marker depths table was used for the subsequent calculations.

To identify markers significantly associated with strain, the *distrib* command was applied. This computed the distribution of markers between lake- and river-strain, noting the marker as present in a given individual if the depth was greater than 5 (*--min-depth 5*). A population map file was inputted to give phenotypic strain assignment to the individuals analyzed. Computation for the *distrib* command includes Pearson's chi-squared test of independence, which calculates the probability of association with the binary trait, and Yates' correction for continuity, which is applied to account for low individual numbers. The cumulative density function of the chi-squared distribution was used to obtain the p-value. The *signif* command was utilized to obtain markers that were significantly associated with strain ($p<0.05$) after Bonferroni correction. Visualization of outputs from the *distrib* and *signif* commands were obtained by the '*sgtr*' R package (Feron 2021). The resulting cladogram distances were computed using the base R dist function and clustering was computed with the base R hclust function (Feron 2021).

The same protocol was followed for analysis of strain phenotype in all individuals except those in RS18 and for the analysis of sex phenotype in all individuals of subgroups RS17_F and RS17_M. The output from these analyses includes the presence or absence of each locus and allele for each individual, allowing for identification of

phenotype-specific markers. The five most significantly associated marker sequences with sufficient coverage were extracted for each analysis. A python script was used to mine for alternative alleles for the significantly associated sequences, or markers (Gamble et al. 2015). The script gives paired reads from the remaining putative phenotype-specific markers that were then uploaded to Geneious (2020.0.3). In Geneious, the loci containing lake-strain and river-strain significantly associated markers were aligned and compared to identify variants. From this identification, primers were designed with the goal of a presence/absence amplification to identify strains.

Results

Analysis of strain phenotype within all individuals except those from subgroup RS18 (n=63) obtained more significant markers than the comparison of strain phenotype between all individuals (n=69). No significant markers were found in the analysis of sex phenotype between known males and females in the RS17 subgroups (n=30). In total, 8,905,875 markers were found in the n=69 analysis, among which 127 sequences were identified as significantly associated with strain (Figures 6 and 7). Within these 127 significant markers, 1 was present in 29 out of 33 lake-strain individuals (Bonferroni corrected p-value=7.91E-06), and 1 was present in 29 out of 36 river-strain individuals (Bonferroni corrected p-value=0.000582). In the n=63 analysis, 7,289,548 markers were found, with 124 sequences significantly associated with strain (Figures 8 and 9). Within these 124 significant markers, 1 was present in 30 out of 33 lake-strain individuals and zero river-strain individuals (Bonferroni corrected p-value=2.36E-05), and 1 was present in 29 out of 30 river-strain individuals that was not present in any lake-strain individuals

(Bonferroni corrected p-value=7.39E-07). In the n=30 analysis, 3,554,054 markers were found but none were significantly associated with sex (Figures 10). Multiple primers were designed around the most significant markers to capture the most variation between strain loci. Of 21 trials, 7 did not work, amplifying in both strains, and 14 did not amplify at all.

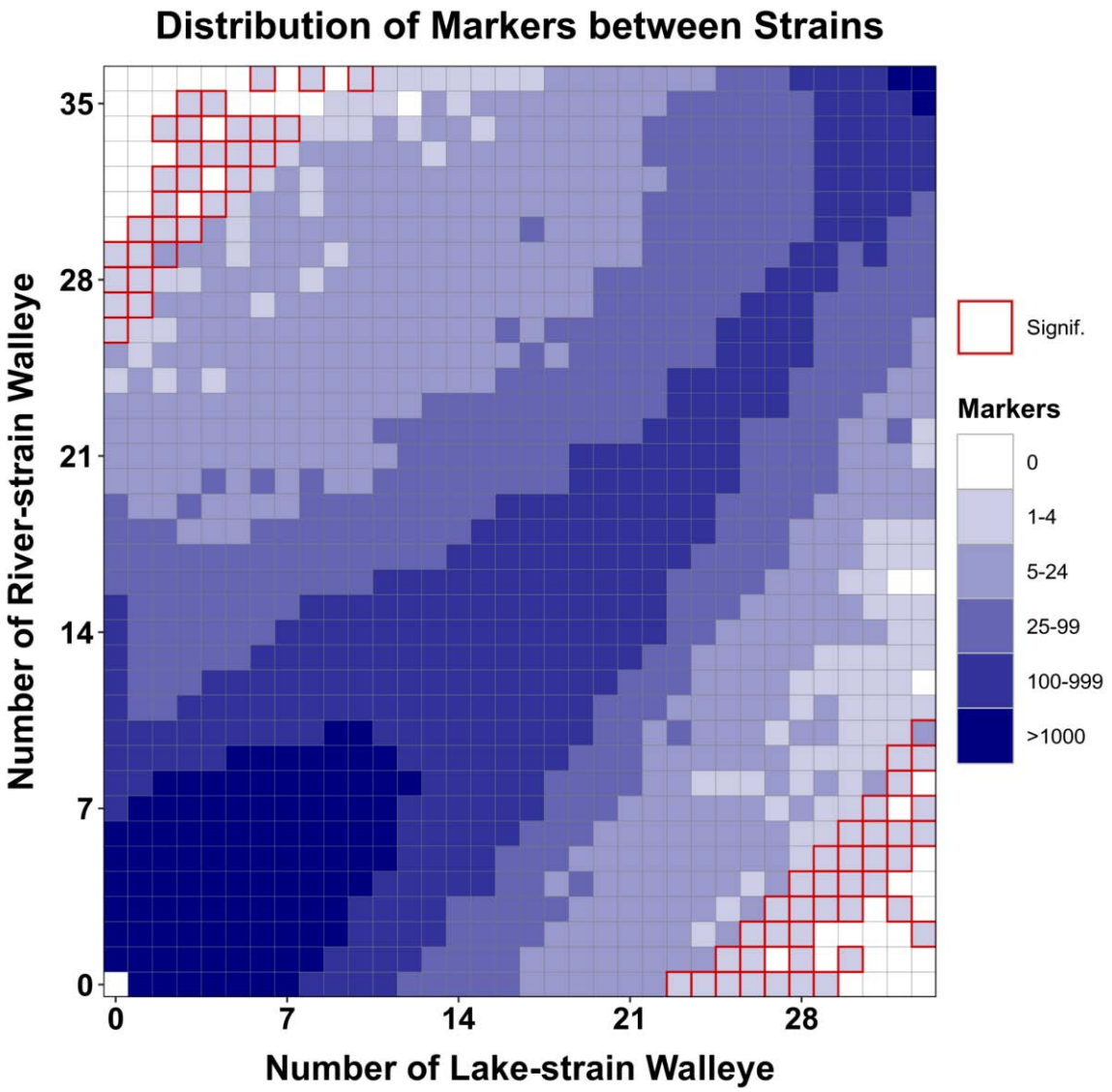

Figure 6. Markers found in the analysis of strain phenotype across all samples (n=69). The tile plot showing distribution of RADSex markers between lake-strain (horizontal axis) and river-strain (vertical axis) using a minimum depth of 5. The color gradient indicates the number of markers present and tiles highlighted with a red border indicate significant association with strain (Chi-squared test, $p < 0.05$ after Bonferroni correction).

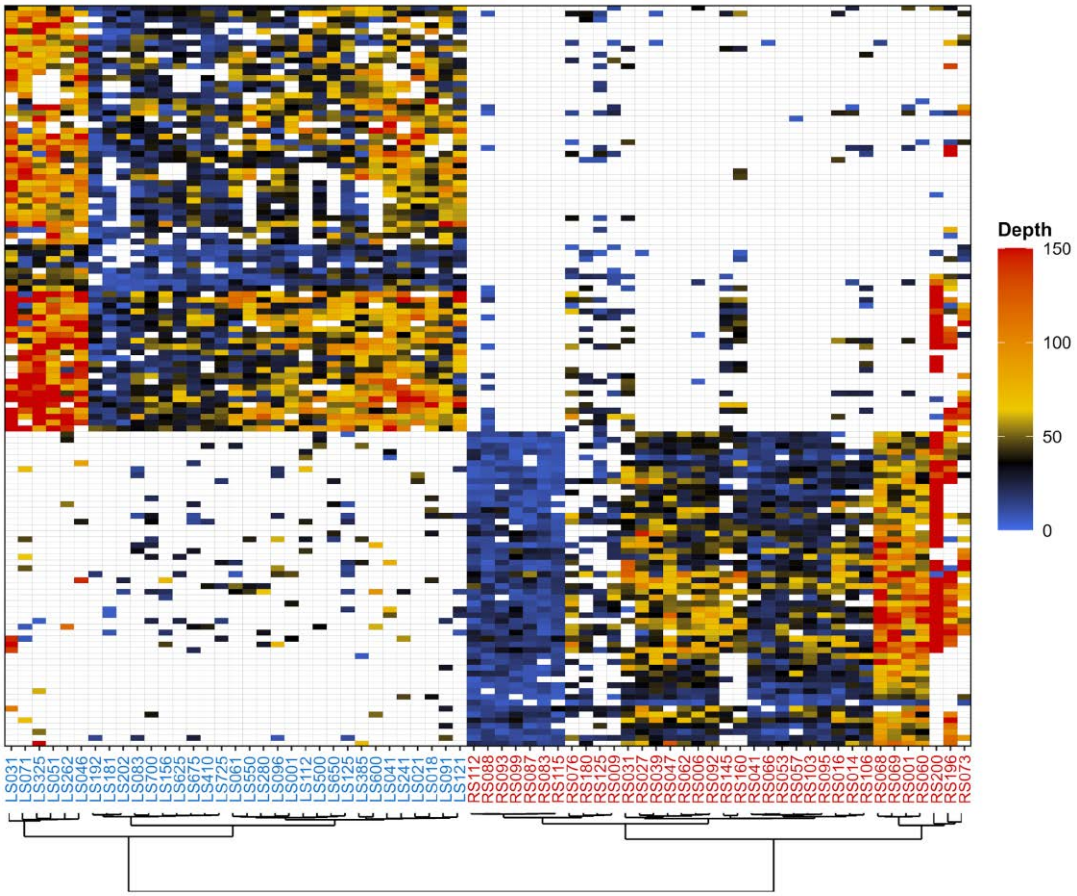

Figure 7. Significant markers found in the analysis of strain phenotype across all samples (n=69). The heatmap showing the sequencing depth of each significant marker (rows) in each individual (columns). The color gradient represents the depth of the marker at each point. Lake-strain individuals (blue labels) and river-strain individuals (red labels) clustered in two separate groups.

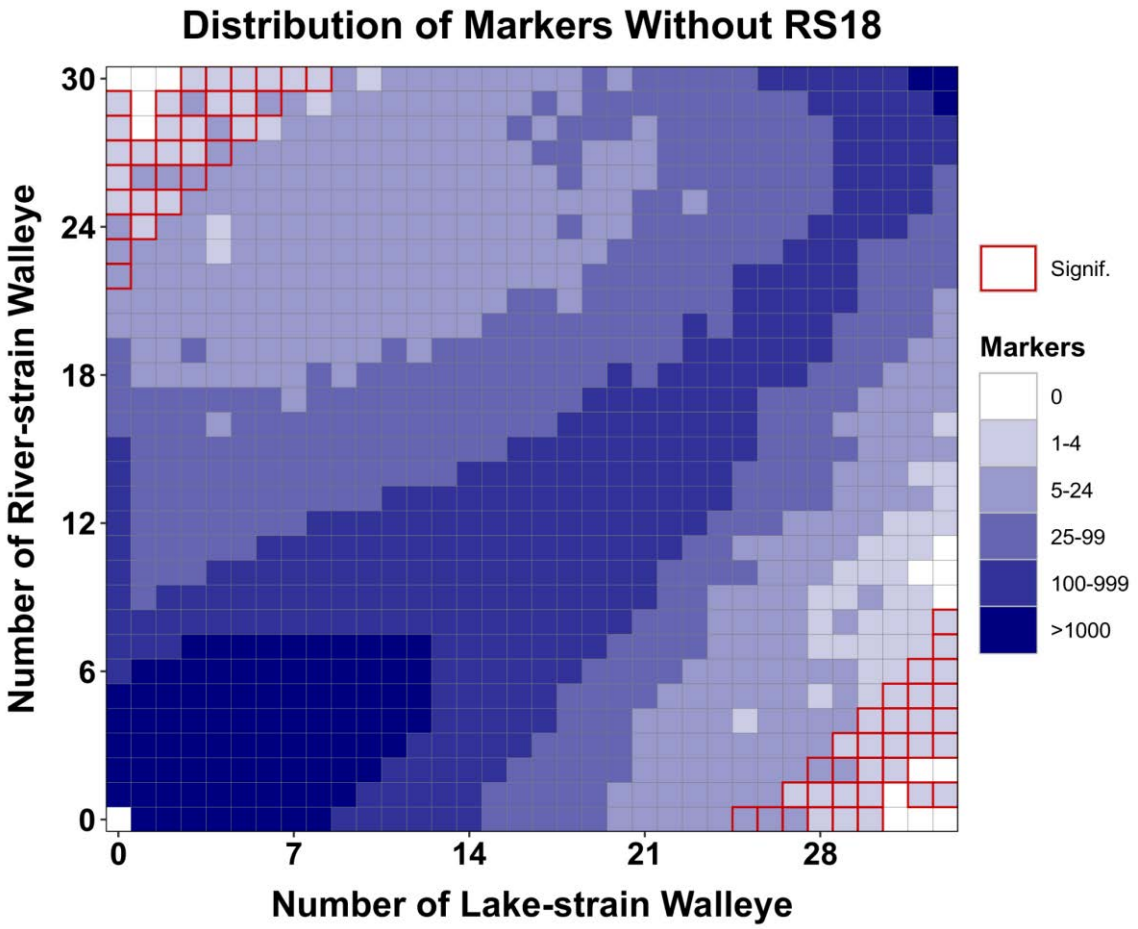

Figure 8. Markers found in the analysis of strain phenotype in all samples excluding the 6 individuals from subgroup RS18 (n=63). The tile plot showing distribution of RADSex markers between lake-strain (horizontal axis) and river-strain (vertical axis) using a minimum depth of 5. The color gradient indicates the number of markers present and tiles highlighted with a red border indicate significant association with strain (Chi-squared test, $p < 0.05$ after Bonferroni correction).

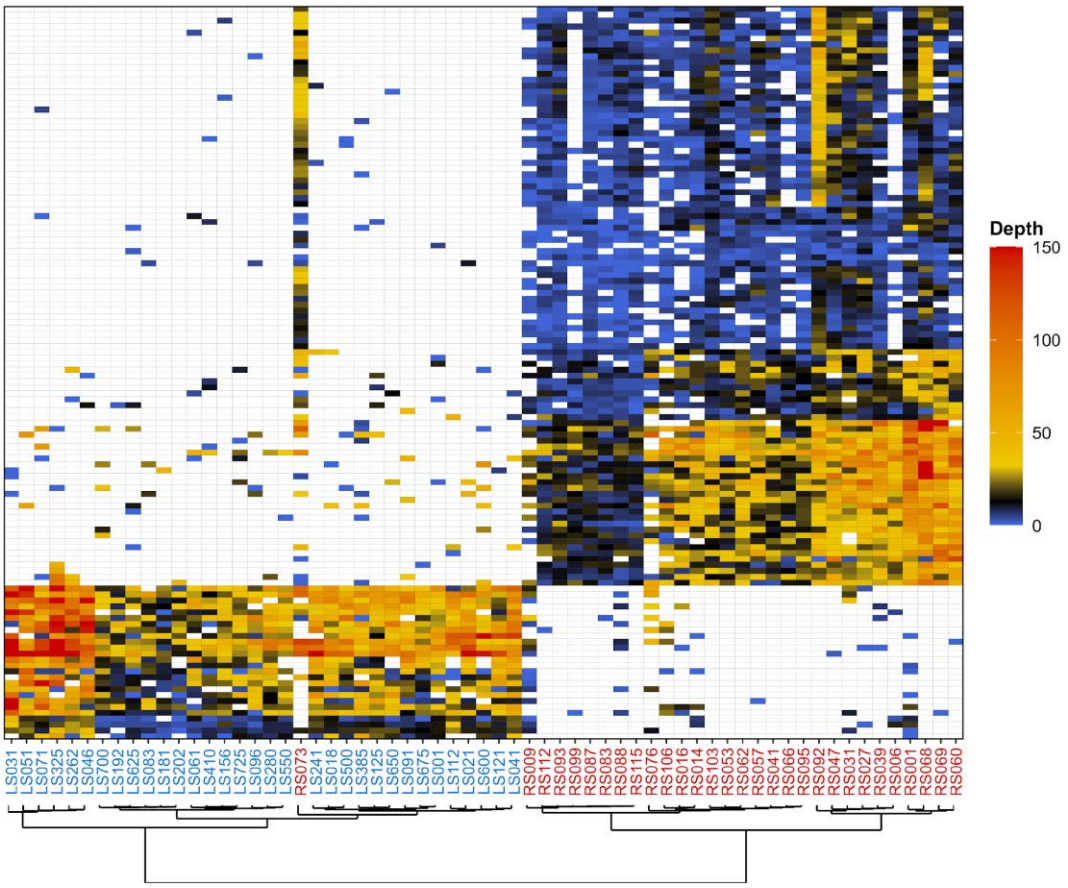

Figure 9. Significant markers found in the analysis of strain phenotype in all samples excluding the 6 individuals from subgroup RS18 (n=63). The heatmap showing the sequencing depth of each significant marker (rows) in each individual (columns). The color gradient represents the depth of the marker at each point. Lake-strain individuals (blue labels) and river-strain individuals (red labels), except for one, clustered in two separate groups with corresponding presence/absence markers.

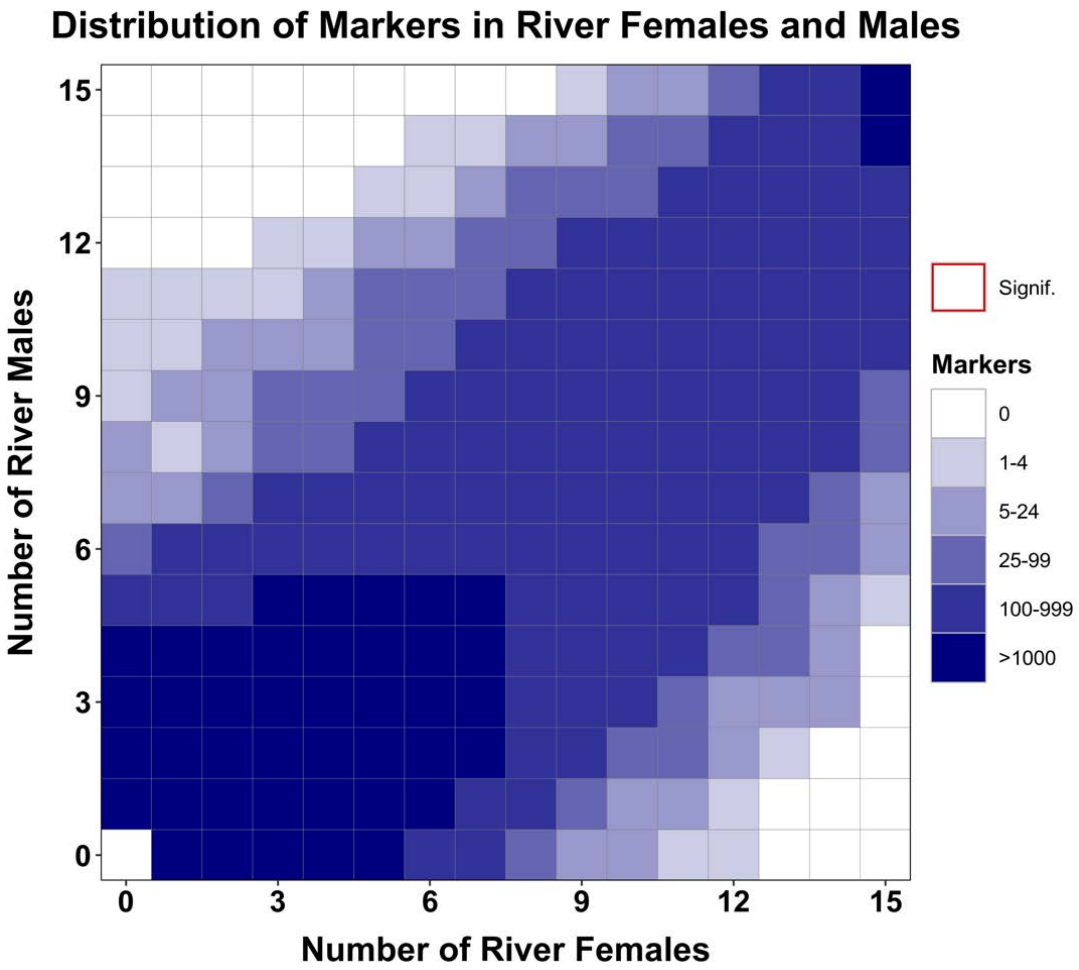

Figure 10. Markers found in the analysis of sex phenotype in subgroups RS17_F and RS17_M (n=30). The tile plot showing distribution of RADSex markers between river-strain females (horizontal axis) and river-strain males (vertical axis). The color gradient indicates the number of markers present. No significant markers were indicated.

Discussion

Multiple markers significantly associated with both lake- and river-strain were found when strain phenotype was compared across all individuals. However, when the individuals from subgroup RS18 were removed from the analysis, the significance was stronger among markers associated with both strains. Removal of the third genetic cluster of individuals allowed for a more in-depth comparison of the two strains. Interestingly,

there were approximately equal numbers of lake-strain associated markers and river-strain associated markers when all the individuals were included, but when the RS18 individuals were removed the proportion shifted identifying about two times more river-strain associated markers than lake-strain associated markers. This reveals that many lake-strain associated markers were coming from differences with RS18 individuals. This indicates that a stock identification tool based off the current data would have varying effectiveness year to year. RADSex was able to identify significantly associated strain markers; however, the difference was minor between strains. Several sequences were outputted as significant, but the base pair difference found when the loci were compared between strains was not enough to differentiate strain and, therefore, create a stock identification tool.

The grouping of individuals, shown by the heatmaps (Figures 7 and 9), were expected, except for the clustering of the phenotypic river-strain individual (RS073) with the phenotypic lake-strain cluster in the n=63 analysis. This could be due to miss identification of the individual's strain at sampling (less likely) or due to its variable haplotype (more likely due to having low to zero coverage for most of the lake-strain associated markers). To better understand the underlying reason, further analysis is underway. Analysis of male and female markers in the RS18 subgroup revealed no significantly associated sex marker, which was supported with findings by Feron et al. (2021).

Conclusion

RADSex has demonstrated an efficient way of handling next generation sequencing data for the identification of significant markers. Significant marker sequences can serve as a building block for genetic methods in stock identification. Advances in genetic methods have made it possible for a single fish scale to be sufficient for DNA extraction and genotyping of the individual (Uglem et al. 2020). Aside from being less invasive, genetic identification methods last the lifetime of the individual and involve less handling time, due to only needing a posteriori assessment. McKinney et al. (2017) is an example of a successful stock identification method with the use of SNP datasets. The identification of strain-specific significant markers is the first step towards a genetic marker for efficient stock identification in walleye.

CHAPTER 5

BROODSTOCK GENETIC CONCLUSIONS

The stocking of *S. vitreus* in Iowa waterbodies is one of the primary tools used by the Iowa DNR to ensure healthy populations for enjoyable fishing experiences. A substantial amount of time and money goes into this effort, so stocking efficiency is vital. Knowledge of genetic diversity contained in the broodstock increases this efficiency by having a better understanding of which strain will survive and contribute to the fishery in certain environments. From the presented data and analyses, genetic groups and ancestry were identified in the Iowa walleye broodstock; however, the genetic signature is subtle and complex. Based on the results of the genetic diversity and marker identification analyses, I recommend that the best tool for identifying stocks would be to generate genome-wide SNP datasets. A full RAD-Seq run on future strain-unknown walleye samples would provide more information on the genetic ancestry of individuals. This would not only allow for the unknown samples to be evaluated alongside the known individuals of this study, revealing genetic strain, but also would give further insight into the ancestry of Iowa walleye by revealing admixture and wild types that were not included in the present broodstock individuals. SNPs are a powerful tool and the valuable information they provide can be applied to optimize stocking. This will improve the walleye fishery in Iowa, ultimately leading to a shorter time between bites.

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